

Matrix discrepancy and optimization

The sharp Lovász-theta constant for random circulant graphs

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: The Fourier-constrained random optimization problem needs sharper analysis than current order bounds; it matters to structured matrix optimization and sampling.

For each $n \geq 2$, partition $\mathbb{Z}_n \setminus \{0\}$ into classes $\{s, -s\}$ (a singleton when $s = -s$). Select each class independently with probability $1/2$ and let S be their union. Form the simple undirected graph G_n on $\mathbb{Z}_n$ with $\{i, j\}$ an edge exactly when $i - j \in S$.

For any graph on n vertices put

$$\vartheta(G) = \max \left\{ \sum_{i,j} X_{ij} : X \in \mathbb{R}^{n \times n}, X = X^T \geq 0, \operatorname{tr} X = 1, X_{ij} = 0 \text{ if } \{i, j\} \in E(G) \right\}.$$

Conjecture. $\lim_{n \rightarrow \infty} \mathbb{E} \vartheta(G_n) / \sqrt{n} = 1$.

Cyclic symmetry reduces this semidefinite program to a linear program involving the discrete Fourier matrix, connecting its typical optimum with structured sampling.

References

1. A. S. Bandeira, D. Dmitriev, K. Lucca, P. Nizić-Nikolac, and A. Rödder, *Randomstrasse101: Open Problems of 2025*, Entry 9, Conjecture 18. [Archived paper](#).
2. A. S. Bandeira, J. Błasiok, D. Dmitriev, U. Faure, A. Kireeva, and D. Kunisky, *The Lovász number of random circulant graphs*, SampTA 2025. [Paper](#); [published DOI](#).

Status check — 2026-09-10

The 2026 collection retains the conjecture and cites the 2025 partial bounds, including an upper bound of order $\sqrt{n \log \log n}$. Searches for subsequent sharp asymptotics and resolutions found none. Independent edge sampling on all vertex pairs is the different ensemble in MD-01.

Audit update (2026-09-10): Rechecked the 2026 Conjecture 18 and the SampTA paper, then searched for sharp random-circulant asymptotics. Added the published DOI; the remaining logarithmic upper-bound factor does not settle the limit. This is a bounded literature check, not a proof that no solution exists.